**D20A / 54**

# EXPERIMENT 9

**Aim:** Create an ERC 20 Token in Remix IDE

## THEORY

## 1. What is a Token?

A token is a digital asset created and managed using a blockchain's smart contract functionality. Tokens can represent currencies, assets or utilities.

Common token standards include:

- ERC-20: Fungible tokens.
- ERC-721: Non-fungible tokens (NFTs).
- ERC-1155: Supports both fungible and non-fungible tokens.

## 2. What is ERC-20?

ERC-20 (Ethereum Request for Comment 20) is a standard for creating fungible tokens on the Ethereum blockchain. Each token is interchangeable with another token of the same type.

It defines common functions and events that allow wallets, exchanges and DApps to interact with tokens.

## 3. Remix IDE

Remix IDE is a browser-based development environment used to write, compile, deploy and test smart contracts written in Solidity.

It provides tools for compiling contracts, connecting wallets and deploying smart contracts on blockchain networks.

## 4. MetaMask

MetaMask is a cryptocurrency wallet used to manage Ethereum accounts and interact with decentralized applications. It connects Remix IDE to the blockchain and is used to approve transactions and pay gas fees.

## 5. Sepolia Testnet

Sepolia is an Ethereum test network used for developing and testing smart contracts. It uses test ETH instead of real ETH to perform transactions and pay gas fees.

## 6. ERC-20 Token Functions

Important ERC-20 functions include:

| Function | Purpose |
|---|---|
| `name()` | Returns token name |
| `symbol()` | Returns token symbol |
| `decimals()` | Returns decimal precision |
| `totalSupply()` | Returns total token supply |
| `balanceOf()` | Checks account balance |
| `transfer()` | Transfers tokens |
| `approve()` | Approves token spending |
| `allowance()` | Checks approved spending limit |
| `transferFrom()` | Transfers approved tokens |

## 7. ERC-20 Events

Events record important activities performed by a smart contract.

- Transfer: Emitted whenever tokens are transferred.
- Approval: Emitted whenever an owner approves another account to spend tokens.

## 8. ERC-20 Attributes

- Name: Full name of the token.
- Symbol: Short representation of the token.
- Decimals: Number of decimal places supported, commonly 18.
- Total Supply: Total number of tokens created.
- Balance: Number of tokens held by an account.

## 9. ERC-20 vs ERC-721

| ERC-20 | ERC-721 |
|---|---|
| Fungible tokens | Non-fungible tokens |
| Tokens are interchangeable | Each token is unique |
| Used for currencies and utility tokens | Used for NFTs |
| No unique ID for each unit | Each token has a unique ID |

## 10. ERC-20 Extensions

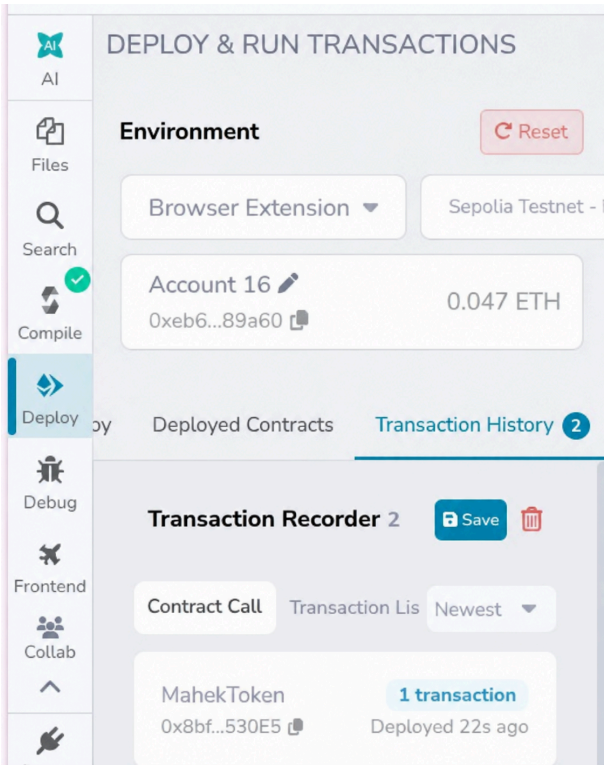
Mintable: Allows creation of new tokens, increasing the total supply.

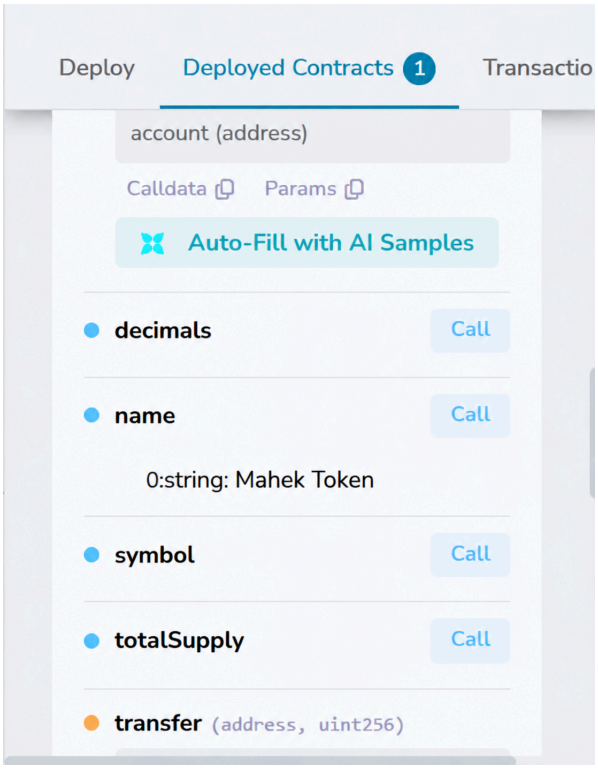
Burnable: Allows permanent destruction of tokens, decreasing the total supply.

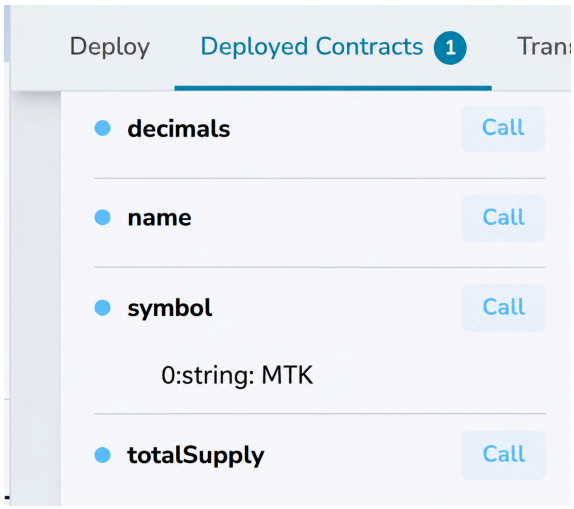
Pausable: Allows authorized accounts to temporarily stop token transfers during emergencies.

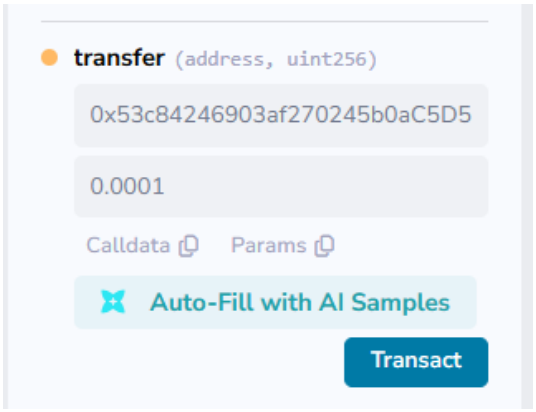
## Output & Code:

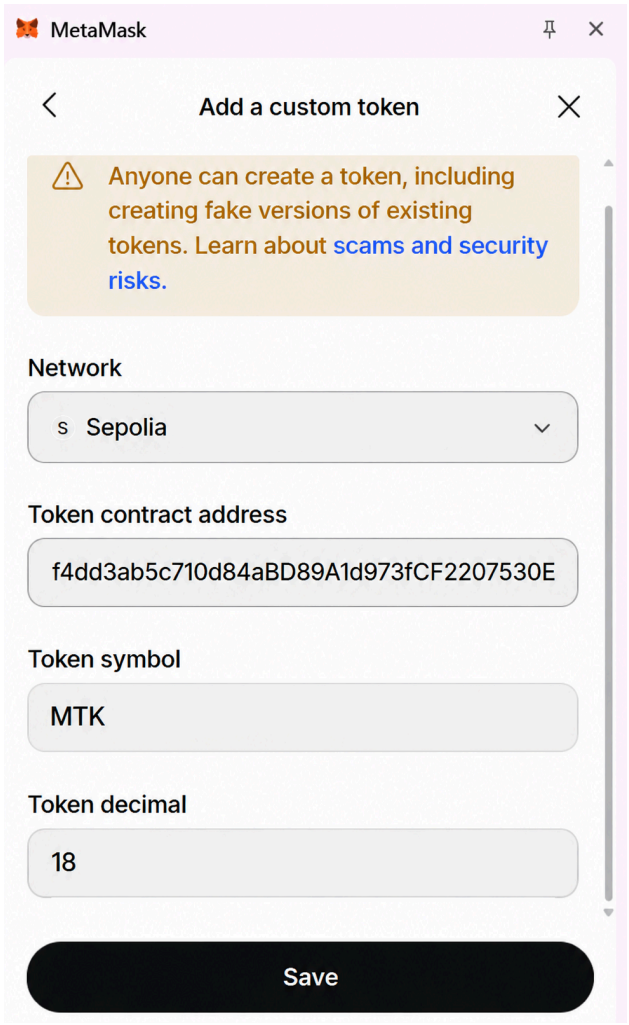
```
1
2   // SPDX-License-Identifier: MIT
3   pragma solidity ^0.8.20;
4
5   import "@openzeppelin/contracts/token/ERC20/ERC20.sol";
6
7   contract MahekToken is ERC20 {
8
9       constructor() ERC20("Mahek Token", "MTK") {     infinite gas 710800 gas
10          _mint(msg.sender, 1000000 * 10 ** decimals());
11      }
12  }
13
```

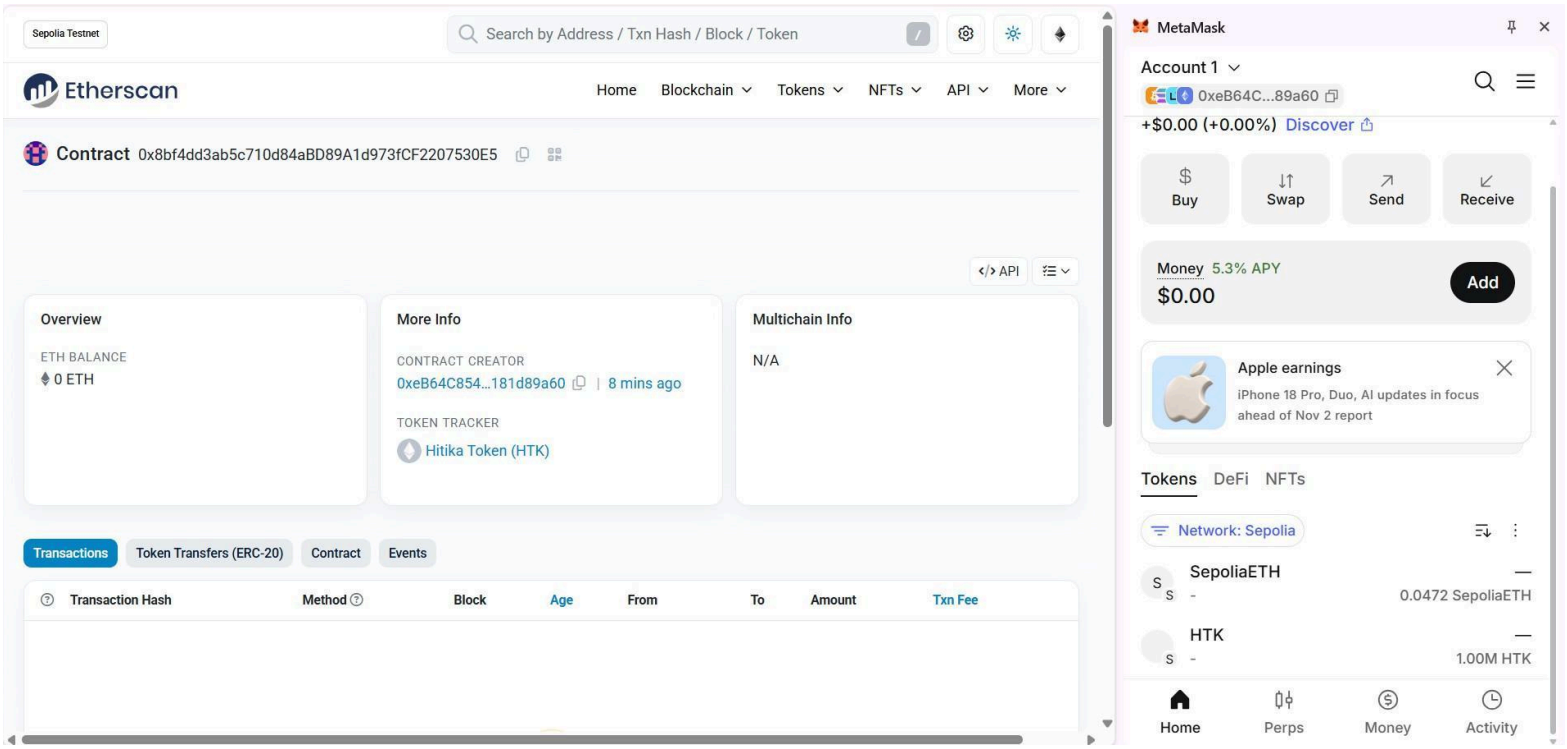



Deploy
Deployed Contracts 1
Transactio
account (address)
Calldata
Params
Auto-Fill with AI Samples
decimals
Call
name
Call
0:string: Mahek Token
symbol
Call
totalSupply
Call
transfer (address, uint256)


Deploy
Deployed Contracts 1
Tran
decimals
Call
name
Call
symbol
Call
0:string: MTK
totalSupply
Call


transfer (address, uint256)
0x53c84246903af270245b0aC5D5
0.0001
Calldata
Params
Auto-Fill with AI Samples
Transact


MetaMask
Add a custom token
Anyone can create a token, including creating fake versions of existing tokens. Learn about scams and security risks.
Network
Sepolia
Token contract address
f4dd3ab5c710d84aBD89A1d973fCF2207530E
Token symbol
MTK
Token decimal
18
Save


Sepolia Testnet
Search by Address / Txn Hash / Block / Token
Etherscan
Home
Blockchain
Tokens
NFTs
API
More
Contract 0x8bf4dd3ab5c710d84aBD89A1d973fCF2207530E5
API
Overview
ETH BALANCE
0 ETH
More Info
CONTRACT CREATOR
0xeB64C854...181d89a60
8 mins ago
TOKEN TRACKER
Hitika Token (HTK)
Multichain Info
N/A
Transactions
Token Transfers (ERC-20)
Contract
Events
Transaction Hash
Method
Block
Age
From
To
Amount
Txn Fee
MetaMask
Account 1
0xeB64C...89a60
+$0.00 (+0.00%)
Discover
Buy
Swap
Send
Receive
Money
5.3% APY
$0.00
Add
Apple earnings
iPhone 18 Pro, Duo, AI updates in focus ahead of Nov 2 report
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
0.0472 SepoliaETH
HTK
1.00M HTK
Home
Perps
Money
Activity